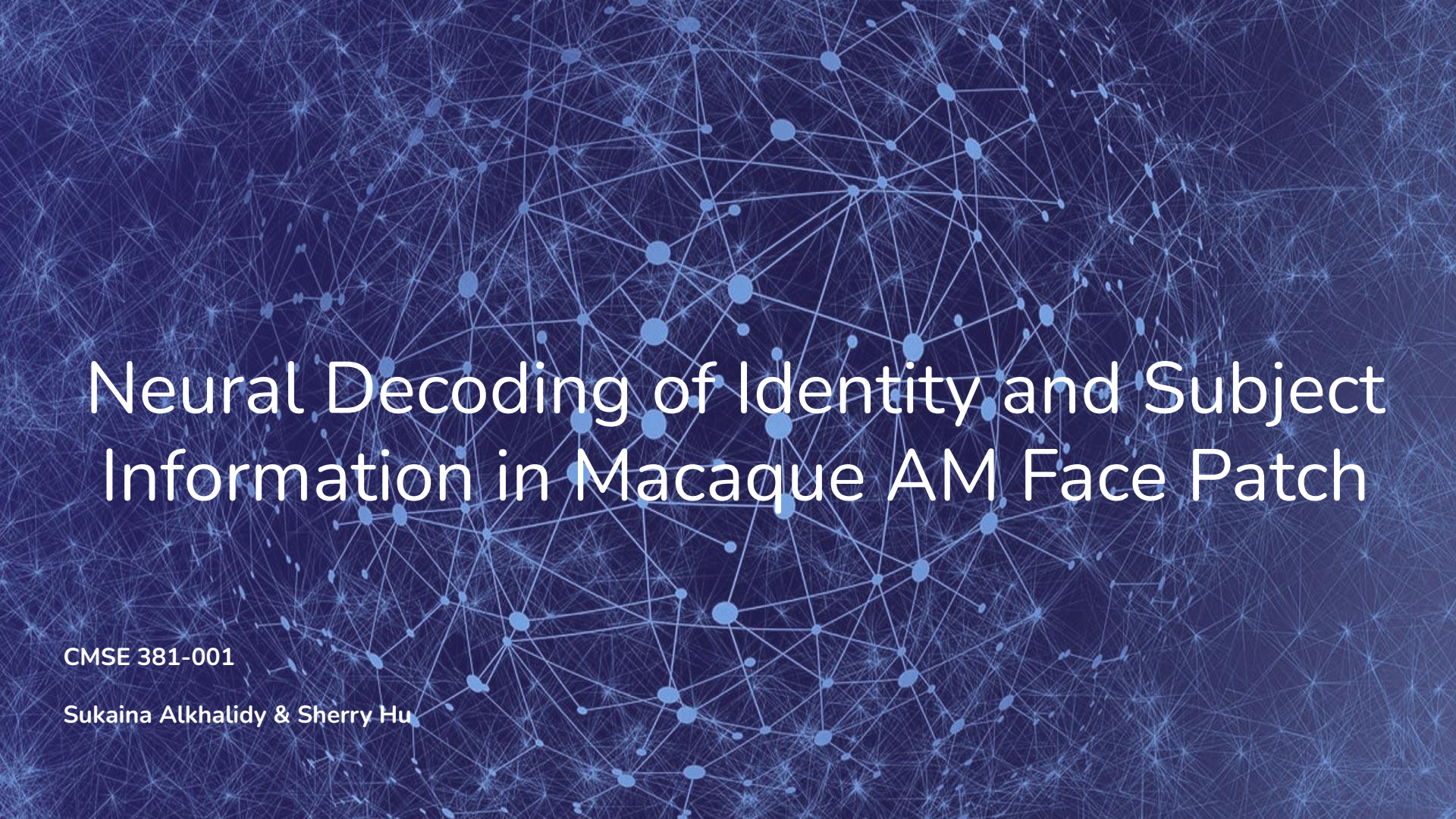




Neural Decoding of Identity and Subject Information in Macaque AM Face Patch

CMSE 381-001

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Background & Motivation

Goal: Investigate what stimulus or subject information can be decoded from neural spike patterns recorded in the macaque Anterior Medial (AM) face patch.

Dataset: Freiwald & Tsao (2010) dataset:

- Neural spike recordings (0/1 per ms) from AM patch.
- Stimuli: 25 face identities at 8 orientations.
- Data combined from 193 files, representing 2 subjects ('bert', 'lupo').

Project Questions:

Q1: Can we classify face identity (1-25) from neural patterns?

Q2: Can we classify the subject ('bert'/'lupo') from neural patterns?

Methodology

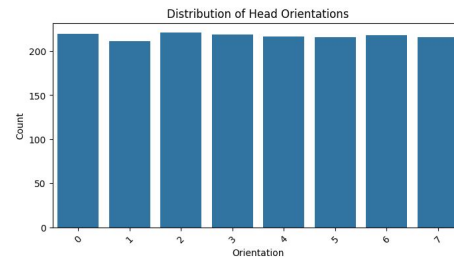
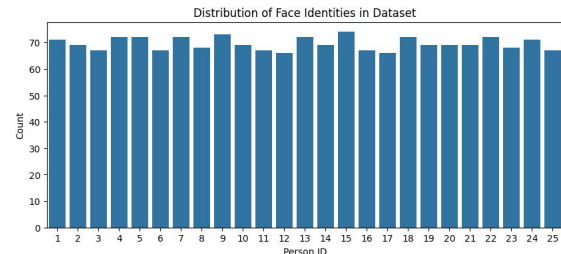
Data Loading: Combined data from 193 CSV files (~206k trials).

Feature Engineering:

- Used neural activity from first 400ms post-stimulus.
- Calculated average firing rates in 8 x 50ms time bins. (Input feature vector for models).
- Well balanced dataset- covers all 25 face identities and 8 orientations evenly
- Hyperparameter testing
- Cross Validation

Modeling:

- Supervised Classification used for both Q1 & Q2.
- Models tested included Logistic Regression, SVC, Random Forest, KNN



Q1: Can we classify face identity (1-25) from neural patterns

Goal: Predict face identity (25 classes)

SVC (Linear Kernel) Accuracy: 0.0833

- Likely underfit due to the linear decision boundary in a complex space
- CV accuracy 0.075

SVC (RBF Kernel) Accuracy: 0.6293

- Captured non-linear patterns in neural activity
- CV accuracy 0.072

K-Nearest Neighbors (KNN) Accuracy: 0.3563

- Struggled with the high dimensionality of neural spike features
- CV accuracy: **0.112**

Random Forest Classifier Accuracy: 0.9770

- Training Accuracy: 97% → clear sign of overfitting
- Robust to noisy, high-dimensional data
- Provided insights via feature importance rankings
- **Random Forest (all)** → CV accuracy: **0.141**

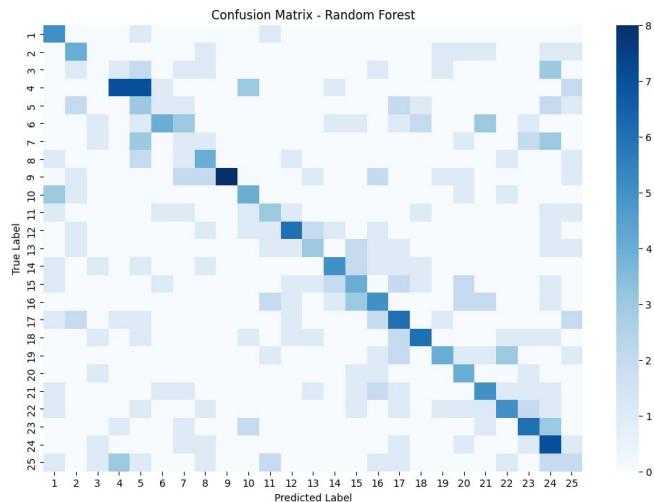
Polynomial Logistic Regression: 0.0201

- Underfit severely, could not capture facial identity patterns.
- Illustrates how complex models is.
- **Not applicable (model failed to learn)**

Adjusted Random Forest (Top 100 Spike-Time Features) : Although the test accuracy dropped to 0.31, the CV accuracy increased to 0.141

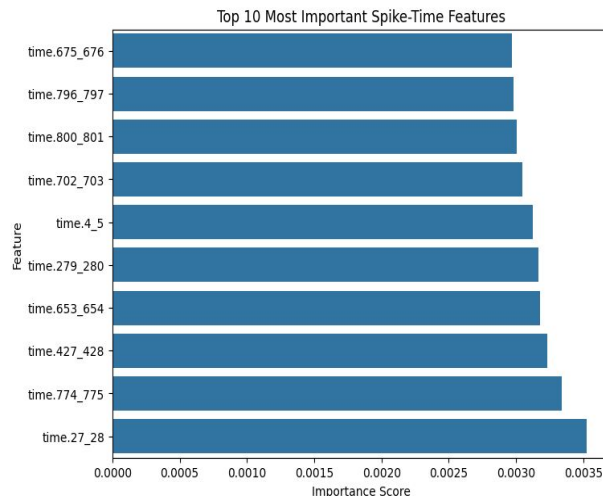
Confusion Matrix

- Shows how well the model predicted each face identity (1–25).
- Diagonal dominance (more values on the diagonal) = more correct predictions.
- Face IDs like **4, 6, 12, 15, 17, 18** were predicted well.
- Some labels (e.g., **10, 24**) had more confusion — suggesting those spike patterns were less distinct.



Top Features:

- Time bins range from **early (e.g., 4–5ms)** to **late (e.g., 800–801 ms)**, showing identity information isn't limited to just early spikes. These top time bins had the highest impact on model decisions, meaning spikes during these millisecond helped distinguish between identities.



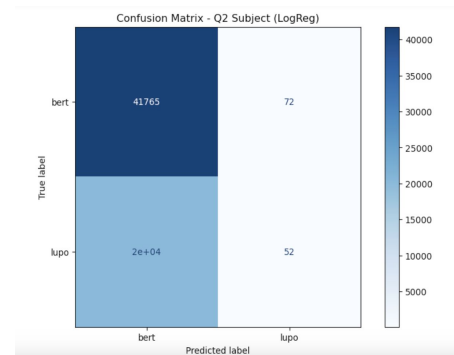
Q2: Classify the subject ('bert' or 'lupo') based on the neural activity patterns

Goal: Predict subject ('bert' vs 'lupo').

Logistic Regression: Test Accuracy: 0.6759

--- Logistic Regression Evaluation (Q2 Subject) ---
Test Accuracy: 0.6759 (Chance Level: 0.50)

Classification Report:				
	precision	recall	f1-score	support
bert	0.68	1.00	0.81	41837
lupo	0.42	0.00	0.01	20028
accuracy			0.68	61865
macro avg	0.55	0.50	0.41	61865
weighted avg	0.59	0.68	0.55	61865



'bert': Recall = 1.00, F1-score = 0.81. The model correctly identifies almost *all* trials that actually belong to 'bert'.

'lupo': Recall = 0.00, F1-score = 0.01. The model identifies practically *none* of the trials that actually belong to 'lupo'. The precision is also low (0.42).

isn't really learning the distinct neural patterns for each subject effectively. Instead, it seems to be primarily predicting the **majority class** ('bert', which has ~41k samples vs 'lupo' with ~20k samples)

Random Forest

Test Accuracy: 0.6764

CV Accuracy score: 0.6772 (k=3)

Classification Report:

	precision	recall	f1-score	support
bert	0.68	0.99	0.81	41837
lupo	0.51	0.02	0.03	20028
accuracy			0.68	61865
macro avg	0.59	0.50	0.42	61865
weighted avg	0.62	0.68	0.55	61865

